MDTS4214_737_4

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2026-02-19

PROBLEM SET 3: Multiple Linear Regression

QUESTION 5.Problem to demonstrate the utility of non- regression over linear regression.

Get the fgl data set from “MASS” library. ***(a)***Considering the refractive index (RI) of “Vehicle Window glass” as the variable of interest and assuming linearity of regression, run multiple linear regression of RI on different metallic oxides. From the p value, report which metallic oxide best explains the refractive index.

```
library(MASS)

data(fgl)
head(fgl)

##      RI    Na   Mg   Al    Si    K   Ca Ba   Fe type
## 1  3.01 13.64 4.49 1.10 71.78 0.06 8.75  0 0.00 WinF
## 2 -0.39 13.89 3.60 1.36 72.73 0.48 7.83  0 0.00 WinF
## 3 -1.82 13.53 3.55 1.54 72.99 0.39 7.78  0 0.00 WinF
## 4 -0.34 13.21 3.69 1.29 72.61 0.57 8.22  0 0.00 WinF
## 5 -0.58 13.27 3.62 1.24 73.08 0.55 8.07  0 0.00 WinF
## 6 -2.04 12.79 3.61 1.62 72.97 0.64 8.07  0 0.26 WinF

#install.packages("stargazer")
library(stargazer)

## Warning: package 'stargazer' was built under R version 4.5.2

##
## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary
Statistics Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

str(fgl)

## 'data.frame':    214 obs. of  10 variables:
##  $ RI  : num  3.01 -0.39 -1.82 -0.34 -0.58 ...
##  $ Na  : num  13.6 13.9 13.5 13.2 13.3 ...
##  $ Mg  : num  4.49 3.6 3.55 3.69 3.62 3.61 3.6 3.61 3.58 3.6 ...
##  $ Al  : num  1.1 1.36 1.54 1.29 1.24 1.62 1.14 1.05 1.37 1.36 ...
```

```
##  $ Si  : num  71.8 72.7 73 72.6 73.1 ...
##  $ K   : num  0.06 0.48 0.39 0.57 0.55 0.64 0.58 0.57 0.56 0.57 ...
##  $ Ca  : num  8.75 7.83 7.78 8.22 8.07 8.07 8.17 8.24 8.3 8.4 ...
##  $ Ba  : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ Fe  : num  0 0 0 0 0 0.26 0 0 0 0.11 ...
##  $ type: Factor w/ 6 levels "WinF","WinNF",..: 1 1 1 1 1 1 1 1 1 1 ...
```

```r
veh <- subset(fgl, type == "Veh")
veh
```

```
##        RI    Na   Mg   Al    Si    K   Ca   Ba   Fe type
## 147 -0.31 13.65 3.66 1.11 72.77 0.11 8.60 0.00 0.00  Veh
## 148 -1.90 13.33 3.53 1.34 72.67 0.56 8.33 0.00 0.00  Veh
## 149 -1.30 13.24 3.57 1.38 72.70 0.56 8.44 0.00 0.10  Veh
## 150 -1.57 12.16 3.52 1.35 72.89 0.57 8.53 0.00 0.00  Veh
## 151 -1.35 13.14 3.45 1.76 72.48 0.60 8.38 0.00 0.17  Veh
## 152  3.27 14.32 3.90 0.83 71.50 0.00 9.49 0.00 0.00  Veh
## 153 -0.21 13.64 3.65 0.65 73.00 0.06 8.93 0.00 0.00  Veh
## 154 -1.90 13.42 3.40 1.22 72.69 0.59 8.32 0.00 0.00  Veh
## 155 -1.06 12.86 3.58 1.31 72.61 0.61 8.79 0.00 0.00  Veh
## 156 -1.54 13.04 3.40 1.26 73.01 0.52 8.58 0.00 0.00  Veh
## 157 -1.45 13.41 3.39 1.28 72.64 0.52 8.65 0.00 0.00  Veh
## 158  3.21 14.03 3.76 0.58 71.79 0.11 9.65 0.00 0.00  Veh
## 159 -0.24 13.53 3.41 1.52 72.04 0.58 8.79 0.00 0.00  Veh
## 160 -0.04 13.50 3.36 1.63 71.94 0.57 8.81 0.00 0.09  Veh
## 161  0.32 13.33 3.34 1.54 72.14 0.56 8.99 0.00 0.00  Veh
## 162  1.34 13.64 3.54 0.75 72.65 0.16 8.89 0.15 0.24  Veh
## 163  4.11 14.19 3.78 0.91 71.36 0.23 9.14 0.00 0.37  Veh
```

```r
model <- lm(RI ~ Na + Mg + Al + Si + K + Ca + Ba + Fe, data = veh)
summary(model)
```

```
##
## Call:
## lm(formula = RI ~ Na + Mg + Al + Si + K + Ca + Ba + Fe, data = veh)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29194 -0.08582  0.00072  0.10740  0.33524
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 131.4641    47.2669   2.781  0.02388 *
## Na           -0.4333     0.3509  -1.235  0.25190
## Mg           -0.2866     1.0075  -0.285  0.78325
## Al           -0.8909     0.5550  -1.605  0.14713
## Si           -1.8824     0.4993  -3.770  0.00547 **
## K            -2.4232     0.9725  -2.492  0.03743 *
## Ca            1.5326     0.5818   2.634  0.02998 *
## Ba            0.3517     2.6904   0.131  0.89922
## Fe            3.8931     0.9581   4.063  0.00362 **
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2621 on 8 degrees of freedom
## Multiple R-squared:  0.9906, Adjusted R-squared:  0.9813
## F-statistic: 105.9 on 8 and 8 DF,  p-value: 2.622e-07
```

```r
stargazer(model,type="text",out="data.txt")
```

```
##
## ===============================================
##                         Dependent variable:
##                     ---------------------------
##                                 RI
## -----------------------------------------------
## Na                           -0.433
##                              (0.351)
##
## Mg                           -0.287
##                              (1.007)
##
## Al                           -0.891
##                              (0.555)
##
## Si                          -1.882***
##                              (0.499)
##
## K                           -2.423**
##                              (0.973)
##
## Ca                           1.533**
##                              (0.582)
##
## Ba                            0.352
##                              (2.690)
##
## Fe                          3.893***
##                              (0.958)
##
## Constant                    131.464**
##                             (47.267)
##
## -----------------------------------------------
## Observations                    17
## R2                             0.991
## Adjusted R2                    0.981
## Residual Std. Error       0.262 (df = 8)
## F Statistic           105.887*** (df = 8; 8)
## ===============================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

```r
summary(model)$coefficients
```

```
##                Estimate Std. Error    t value    Pr(>|t|)
## (Intercept) 131.4640676 47.2669236  2.7813121 0.023876172
## Na           -0.4333080  0.3508773 -1.2349274 0.251895370
## Mg           -0.2866243  1.0074637 -0.2845009 0.783251988
## Al           -0.8908690  0.5550086 -1.6051446 0.147129402
## Si           -1.8823864  0.4993058 -3.7700067 0.005465591
## K            -2.4231984  0.9725295 -2.4916451 0.037426154
## Ca            1.5326244  0.5817872  2.6343387 0.029975590
## Ba            0.3517015  2.6904136  0.1307240 0.899221141
## Fe            3.8931318  0.9580806  4.0634699 0.003616000
```

```r
#remove intercept and find smallest p value
pvals=summary(model)$coefficients[-1,4]
pvals
```

```
##          Na          Mg          Al          Si           K          Ca
## 0.251895370 0.783251988 0.147129402 0.005465591 0.037426154 0.029975590
##          Ba          Fe
## 0.899221141 0.003616000
```

```r
best_predictor=names(which.min(pvals))
best_predictor
```

```
## [1] "Fe"
```

Conclusion The metallic oxide with the smallest p-value is: Fe :Iron It best explains RI under linear regression assumption.

(b) Run a simple linear regression of RI on the best predictor chosen in (a).

```r
# Create formula dynamically
formula_simple = as.formula(paste("RI ~", best_predictor))

# Fit simple linear regression
model_simple = lm(formula_simple, data = veh)

# Summary
summary(model_simple)
```

```
##
## Call:
## lm(formula = formula_simple, data = veh)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.2324 -1.0693 -0.2715  0.2907  3.7707
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  -0.5007     0.4861  -1.030   0.3193
## Fe            8.1362     4.0780   1.995   0.0645 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.759 on 15 degrees of freedom
## Multiple R-squared:  0.2097, Adjusted R-squared:  0.157
## F-statistic: 3.981 on 1 and 15 DF,  p-value: 0.06452
```

```r
summary(model_simple)$r.squared
```

```
## [1] 0.2097192
```

Conclusion:

*RI= -0.5007 + 8.1362 * Fe*

Intercept (β_0) = −0.5007

Fe (β_1) = 8.1362 ; p-value = 0.06452

This indicates that for every one-unit increase in Fe content, the refractive index (RI) increases on average by 8.1362 units.

The model explains 20.97% of the variability in refractive index.

The predictor Fe has a p-value of 0.0645, which is:

1. Not significant at the 5% level

2. Marginally significant at the 10% level

(c)Can you further improve the regression of the refractive index of "Vehicle Window glass" on the predictor chosen by you in part (a)? Give the new fitted model and compare its performance with the model in (b).

```r
# Simple linear regression (already fitted earlier, but refitting for
clarity)
model_simple = lm(RI ~ Fe, data = veh)

summary(model_simple)
```

```
##
## Call:
## lm(formula = RI ~ Fe, data = veh)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.2324 -1.0693 -0.2715  0.2907  3.7707
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5007     0.4861  -1.030   0.3193
## Fe            8.1362     4.0780   1.995   0.0645 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 1.759 on 15 degrees of freedom
## Multiple R-squared:  0.2097, Adjusted R-squared:  0.157
## F-statistic: 3.981 on 1 and 15 DF,  p-value: 0.06452
```

```r
# Store performance metrics
r2_simple = summary(model_simple)$r.squared
adjr2_simple = summary(model_simple)$adj.r.squared
```

```r
# Quadratic regression model
model_quad = lm(RI ~ Fe + I(Fe^2), data = veh)

summary(model_quad)
```

```
##
## Call:
## lm(formula = RI ~ Fe + I(Fe^2), data = veh)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.6215 -1.1715 -0.1345  0.5985  3.5485
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.2785     0.4712  -0.591    0.564
## Fe          -12.1810    12.0408  -1.012    0.329
## I(Fe^2)      65.9600    37.0798   1.779    0.097 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.645 on 14 degrees of freedom
## Multiple R-squared:  0.3554, Adjusted R-squared:  0.2633
## F-statistic:  3.86 on 2 and 14 DF,  p-value: 0.04623
```

```r
# Store performance metrics
r2_quad = summary(model_quad)$r.squared
adjr2_quad = summary(model_quad)$adj.r.squared
```

```r
# Compare R-squared
r2_simple
```

```
## [1] 0.2097192
```

```r
r2_quad
```

```
## [1] 0.355413
```

```r
# Compare Adjusted R-squared
adjr2_simple
```

```
## [1] 0.1570338
```

```
adjr2_quad

## [1] 0.2633292
```

The quadratic model improves the regression if: R^2 (quadratic) > R^2 (simple) Adjusted R^2 increases The quadratic term is marginally significant at 10% level (p = 0.097), suggesting that nonlinear regression provides a slight improvement over the simple linear model,though the improvement is not strong at the 5% significance level.

PROBLEM SET 4: Some Potential Problems in Multiple Linear Regression

QUESTION 1:***Problem to demonstrate multicollinearity***

Consider the Credit data in the ISLR library.

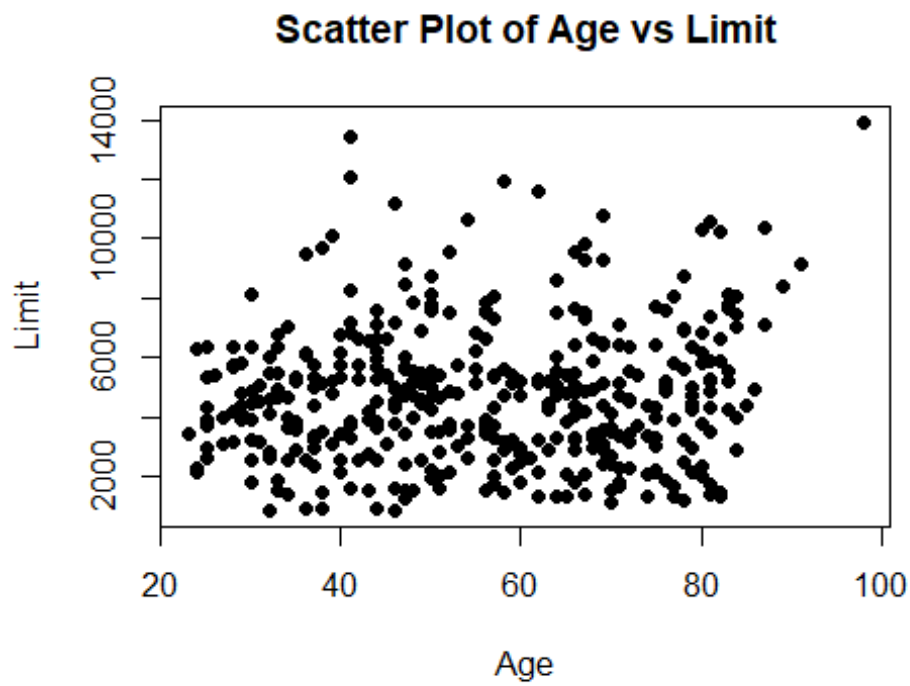
Choose balance as the response and Age, Limit and Rating as the predictors.

```
library(ISLR)
attach(Credit)
df=data.frame(Balance,Age,Limit,Rating)
attach(df)

## The following objects are masked from Credit:
##
##     Age, Balance, Limit, Rating
```

(a) Make a scatter plot of

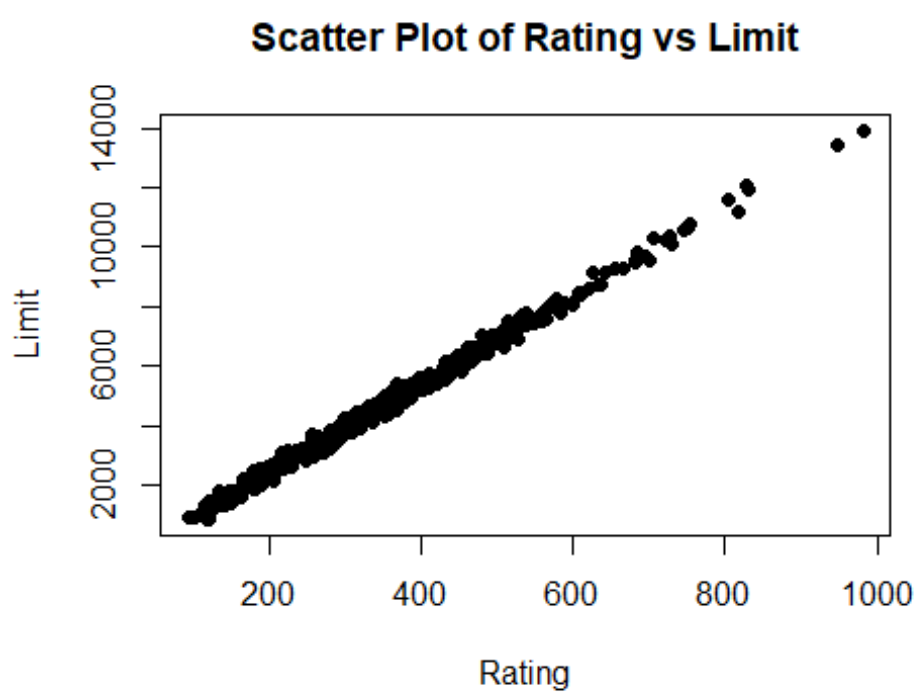
(i) Age versus Limit and

```
plot(x=Age,y=Limit,type="p",pch=16,main="Scatter Plot of Age vs Limit")
```



(ii) Rating Versus Limit.

```
plot(x=Rating,y=Limit,type="p",pch=16,main="Scatter Plot of Rating vs Limit")
```



Comment on the scatter plot.

Comment on respective Scatter Plots:

(i)Age versus Limit

*1.*Type:Linear

*2.*Direction:Positive(As Age rises Limit rises on an average)

*3.*Strength:Weak(Points on the scatter plot are well dispersed)

(ii)Rating Versus Limit

*1.*Type:Linear

*2.*Direction:Positive(As Rating rises Limit rises on an average)

*3.*Strength:Strong(Points on the scatter plot are very closely placed)

Overall Comment:

In terms of Collinearity-

*1.*Age and Limit having weak strength of association may not be collinear. *2.*Rating and Limit having strong strength of association are suspected to be highly collinear.

(b) Run three separate regressions:

(i) Balance on Age and Limit

```
m1=lm(Balance~Age+Limit,data=df)
```

(ii) Balance on Age, Rating and Limit

```
m2=lm(Balance~Age+Rating+Limit,data=df)
```

(iii) Balance on Rating and Limit.

```
m3=lm(Balance~Rating+Limit,data=df)
```

Present all theregression output in a single table using stargazer.

```
library(stargazer)
stargazer(m1,m2,m3,type='text',out='f1.txt')

##
##
==============================================================================
=================
##                                                Dependent variable:
##                      -------------------------------------------------------
--------------------
##                                                          Balance
```

```
##                                (1)                      (2)
(3)
## ---------------------------------------------------------------------------
--------------------
## Age                        -2.291***                -2.346***
##                             (0.672)                  (0.669)
##
## Rating                                               2.310**
2.202**
##                                                      (0.940)
(0.952)
##
## Limit                       0.173***                  0.019
0.025
##                             (0.005)                  (0.063)
(0.064)
##
## Constant                  -173.411***              -259.518***
-377.537***
##                             (43.828)                 (55.882)
(45.254)
##
## ---------------------------------------------------------------------------
--------------------
## Observations                  400                      400
400
## R2                           0.750                    0.754
0.746
## Adjusted R2                  0.749                    0.752
0.745
## Residual Std. Error    230.532 (df = 397)       229.080 (df = 396)
232.320 (df = 397)
## F Statistic         594.988*** (df = 2; 397) 403.718*** (df = 3; 396)
582.820*** (df = 2; 397)
##
===========================================================================
=================
## Note:                                                          *p<0.1;
**p<0.05; ***p<0.01
```

What is the marked difference that you can observe from the output?

Comment:

As mentioned previously this output justifies the overall comment in part (a).

The standard error of the coefficient estimate of Limit Predictor was-

*1.*Lowest(=0.005,very low) while inclusion of Age as also a predictor.

*2.*Highest(=0.064,drastically high) while inclusion of Rating as also a predictor.

*3.*Medium but not very low(=0.063) while inclusion of Age and Rating as predictors.

This strongly justifies that Predictors Age and Limit are not collinear while Rating and Limit are collinear.

(c)Calculate the variance inflation factor (VIF) and comment on multicollinearity.

```
#Install car Package
library(car)

## Warning: package 'car' was built under R version 4.5.2

## Loading required package: carData

## Warning: package 'carData' was built under R version 4.5.2

vif(m1)

##      Age    Limit
## 1.010283 1.010283

vif(m2)

##        Age     Rating      Limit
##   1.011385 160.668301 160.592880

vif(m3)

##   Rating    Limit
## 160.4933 160.4933
```

Comment: The Variance Inflation factor of models-

1.'m1' after regressing Age on Limit and vice-versa does not exceed 5 or 10.

2.'m3' after regressing Rating on Limit and vice-versa exceeds 10.

3.'m2' after regressing -

Age on Rating and Limit does not exceeds 5 or 10.

Rating on Age and Limit exceeds 10.

Limit on Rating and Limit exceeds 10.

Thus by thumb rule, Age is not collinear with any other predictors but Limit and Rating are Collinear.

QUESTION 2:*** Problem to demonstrate the detection of outlier, leverage and influential points***

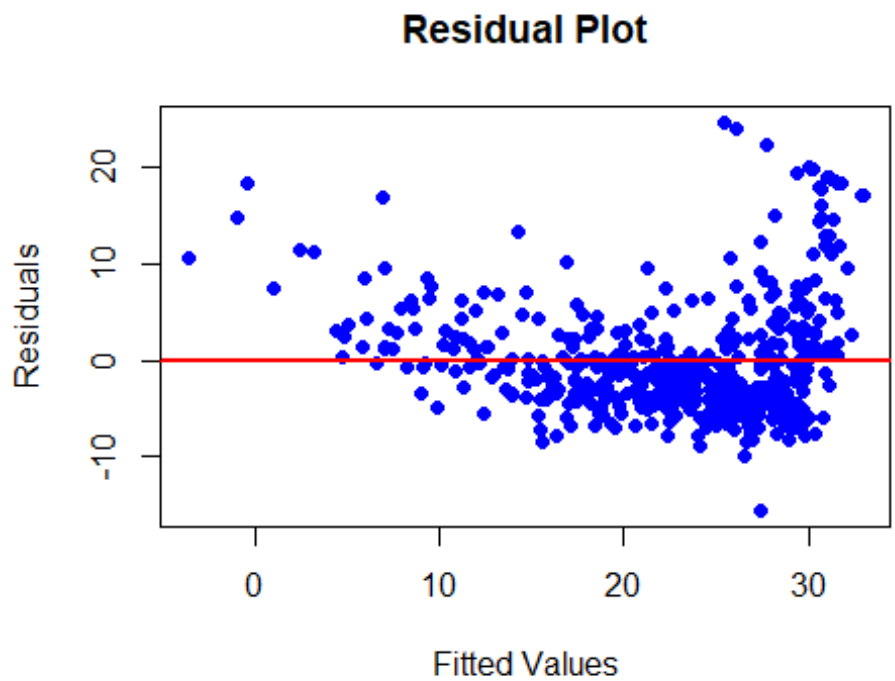
Attach "Boston" data from MASS library in R.

Select median value of owner-occupied homes, as the response and per capita crime rate, nitrogen oxides concentration, proportion of blacks and percentage of lower status of the population as predictors.

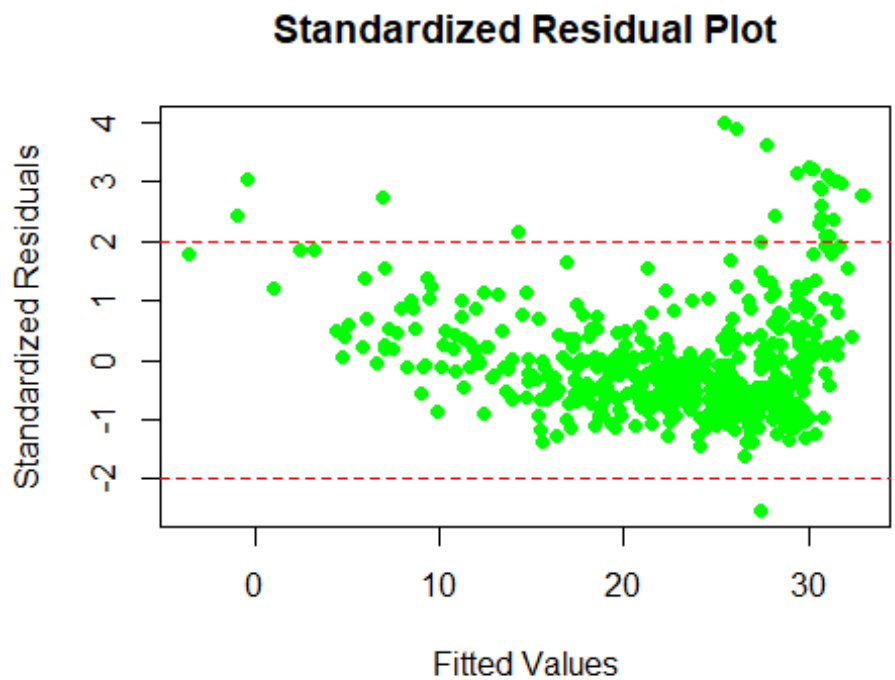
```
library(MASS)
attach(Boston)
df=data.frame(medv,crim,nox,black,lstat)
attach(df)

## The following objects are masked from Boston:
##
##     black, crim, lstat, medv, nox
```

The objective is to fit a multiple linear regression model of the response on the predictors.

```
mlr<- lm(medv ~ crim + nox + black + lstat, data = df)
summary(mlr) # Model summary

##
## Call:
## lm(formula = medv ~ crim + nox + black + lstat, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -15.564  -4.004  -1.504   2.178  24.608
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 30.053584   2.170839  13.844   <2e-16 ***
## crim        -0.059424   0.037755  -1.574    0.116
## nox          3.415809   3.056602   1.118    0.264
## black        0.006785   0.003408   1.991    0.047 *
## lstat       -0.918431   0.050167 -18.307   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.183 on 501 degrees of freedom
## Multiple R-squared:  0.5517, Adjusted R-squared:  0.5481
## F-statistic: 154.1 on 4 and 501 DF,  p-value: < 2.2e-16

plot(mlr$fitted.values, resid(mlr),
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Residual Plot",
     pch = 19, col = "blue")

abline(h = 0, col = "red", lwd = 2)
```



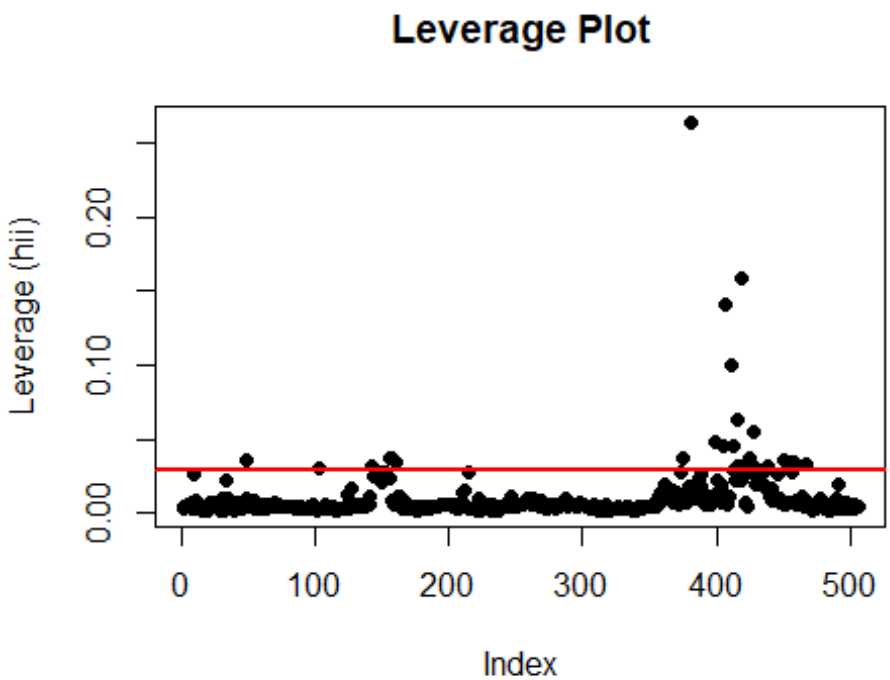

```
s_resid <- rstandard(mlr) # Standardized residuals

# Plot standardized residuals
plot(mlr$fitted.values, s_resid,
     xlab = "Fitted Values",
     ylab = "Standardized Residuals",
     main = "Standardized Residual Plot",
     pch = 16, col = "green")

abline(h = c(-2, 2), col = "red", lty = 2)
```




```
which(abs(s_resid) > 2.5)  # Observations that may be outliers

## 162 163 164 167 187 196 204 205 215 226 229 234 258 263 268 284 369 370
371 372
## 162 163 164 167 187 196 204 205 215 226 229 234 258 263 268 284 369 370
371 372
## 373 413 506
## 373 413 506
```

Leverage points

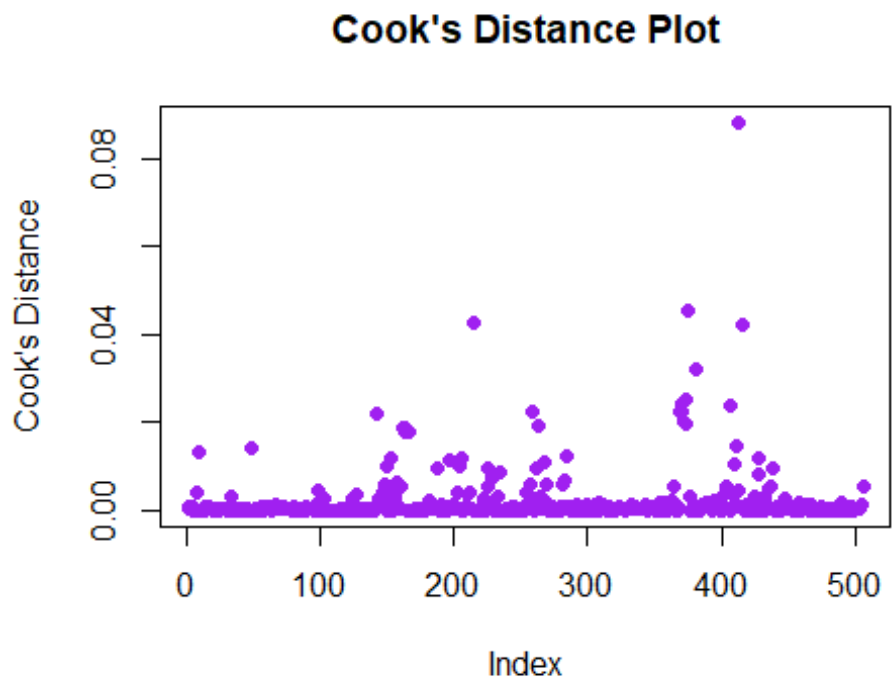
```
# Leverage (hat values)
hii = hatvalues(mlr)

# Plot leverage
plot(hii,
     ylab = "Leverage (hii)",
     main = "Leverage Plot",
     pch = 16)

# Cutoff: 3*(p+1)/n
n = nrow(Boston)
p = length(coef(mlr)) - 1
cutoff = 3 * (p + 1) / n
abline(h = cutoff, col = "red", lwd = 2)
```




```
# Observations with high leverage
which(hii > cutoff)

##  49 103 142 156 157 160 375 381 399 405 406 411 413 415 416 417 419 424
425 426
##  49 103 142 156 157 160 375 381 399 405 406 411 413 415 416 417 419 424
425 426
## 427 428 438 439 451 455 457 458 467
## 427 428 438 439 451 455 457 458 467
```

Influential points

```
# Cook's distance
cooksd = cooks.distance(mlr)

# Plot Cook's distance
plot(cooksd,
     ylab = "Cook's Distance",
     main = "Cook's Distance Plot",
     pch = 16, col = "purple")

abline(h = 1, col = "black", lty = 2)
```




```r
# Identify influential points
which(cooksd > 1)

## named integer(0)
```